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AI and IoT-Based Semi-Autonomous Firefighting Drone

(A Smart and Sustainable Solution for Industrial
Safety)

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Abstract

Unmanned Aerial Vehicles (UAVs), or drones, are increasingly used to perform critical tasks in hazardous environments, reducing human risk—especially in firefighting and inspection work. This project focuses on developing a cost-effective firefighting drone equipped with flame detection sensors, temperature sensors, and a live camera feed to identify and suppress fires in hard-to-reach areas.

The drone integrates a mini submersible water pump connected to a 300–500 mL tank, allowing it to discharge water or extinguishing liquid directly over small-scale fires. It is built on a lightweight carbon fiber frame with brushless motors, electronic speed controllers, and a LiPo battery, ensuring stability and durability during flight. Control is handled through a FlySky FS-i6 transmitter and ESP32-CAM module, which provides live video monitoring and basic IoT connectivity.

This project demonstrates how a simple, low-cost UAV can assist firefighters by reaching dangerous zones, performing flame detection, and initiating early suppression — reducing both response time and risk to human life.

Introduction

With rapid advancements in technology, drones have become powerful tools for performing tasks that are too dangerous or inaccessible for humans. Their ability to navigate challenging locations makes them especially valuable in firefighting and inspection work—where safety and speed are critical.

Fire incidents often occur in areas where human access is limited, such as high-rise buildings, industrial zones, or dense forests. Traditional firefighting methods rely heavily on human effort, exposing firefighters to extreme heat, toxic fumes, and unpredictable conditions. This project explores the use of drones as an assistive technology to minimize these risks by detecting and suppressing fires remotely.

The developed drone integrates basic electronic and IoT systems to enable early fire detection and water-based suppression. It employs IR flame sensors to identify fire sources, a temperature sensor to confirm heat levels, and an ESP32-CAM module to provide real-time visual feedback. Upon detecting a fire, the system activates a mini water pump that sprays water from an onboard tank onto the affected area.

Unlike advanced firefighting drones equipped with heavy extinguishing systems or GPS navigation, this design prioritizes a lightweight, low-cost, and educational approach. It is intended for handling smaller fire incidents or serving as a prototype for further research and development. The project highlights how affordable hardware and simple automation can improve response time, enhance safety, and reduce human involvement in hazardous environments.

Problem Statement

Traditional firefighting methods are inherently risky, labor-intensive, and often ineffective in handling fires that occur in difficult-to-access or hazardous locations. Firefighters frequently risk their lives while operating under extreme heat, smoke, and structural instability.

This project addresses the following key challenges:

Inaccessibility: Conventional firefighting equipment and personnel face limitations in reaching fires in high-rise buildings, rugged terrains, or dense forests. Such inaccessibility increases the potential for greater damage and structural failure.

Safety Concerns: Fire incidents expose responders to severe dangers such as heat exhaustion, burns, toxic fumes, and high psychological stress.

Response Delay: Fire accidents occur suddenly and demand immediate action. Delays caused by traffic, equipment setup, or manual response can allow the fire to spread rapidly, leading to significant losses.

To mitigate these challenges, this project proposes a lightweight, IoT-enabled drone capable of detecting and suppressing fires remotely. The system aims to reduce human involvement in hazardous environments, improve response time, and minimize both human casualties and property damage

Objective

The primary goal is to develop a versatile aerial system, designing a robust UAV (Quadcopter or Hexacopter) platform capable of carrying specific payloads for fire suppression, while potentially incorporating additional functions such as surveillance. The project also aims to enhance fire detection and monitoring by integrating advanced sensors, including thermal

cameras, GPS tracking, and high-resolution imaging, to improve situational awareness, localize fire hotspots, and monitor the extent of the fire in real-time. Another key objective is to implement an efficient extinguishing mechanism, creating a system that can rapidly deploy extinguishing agents (such as fire extinguisher balls or liquid spray) to the fire source, especially in hard-to-reach areas. Additionally, the project seeks to maximize system reliability and stability, ensuring the drone remains stable during flight, even under challenging conditions and with varying payloads. Finally, the system aims to minimize human risk, enabling remote operation to contain or extinguish fires without requiring human presence in hazardous zones.

Scope of the Study / Project

This project focuses on creating a semi-autonomous firefighting drone that can detect and respond to fire-related situations using infrared sensors, IoT connectivity, and manual human control. The aim is to design a system that is simple, affordable, and effective enough to be used in real-world industrial environments such as warehouses, factories, and logistics centers, where even small fires can cause serious damage.

The drone will detect unusual heat patterns using an infrared sensor and stream that data to a laptop-based control system. A human pilot will then guide the drone through a remote controller, allowing it to aim and spray water through a lightweight pipe connected to a ground tank. This approach ensures that human judgment and control remain central while still using technology to make firefighting faster and safer.

The project mainly focuses on building a working prototype using low-cost and easily available components such as 1000KV motors, Botwing F405 flight controller, IR sensors, and a 433 MHz RF relay system. The design emphasizes practicality over complexity, proving that meaningful innovation can be achieved without expensive hardware.

In the long run, this project can be scaled and improved with AI-based fire detection, GPS navigation, and autonomous flight capabilities, making it a part of future smart factory and industrial safety systems. It also supports global sustainability efforts like the UN Sustainable Development Goals — SDG 9 (Industry, Innovation, and Infrastructure) and SDG 13 (Climate Action) by promoting safer, more responsible technology use.

Overall, the scope of this project is to show how a blend of human control, smart sensors, and IoT data can lead to an efficient and sustainable solution for real-world fire emergencies — all while staying accessible to students, startups, and small industries.

Literature Review

In recent years, the use of Unmanned Aerial Vehicles (UAVs) in industrial and emergency applications has grown significantly, particularly in the fields of fire detection and infrastructure inspection. Drones have proven to be effective in improving safety and accessibility in hazardous environments by integrating sensors and IoT technologies that enable autonomous monitoring and response.

For fire detection, UAVs equipped with temperature and flame sensors have demonstrated high accuracy in identifying potential fire outbreaks, minimizing the need for human intervention in dangerous conditions. Similarly, in bridge inspection, drones fitted with high-resolution cameras can capture detailed images and video footage of bridge components, helping detect structural damage such as cracks and corrosion more safely and efficiently than manual inspection methods.

While advanced UAV systems often incorporate GPS navigation, thermal cameras, and heavy payloads, these solutions tend to be expensive and complex. Hence, this project focuses on a cost-effective, lightweight prototype utilizing an ESP32-CAM for live video transmission, IR flame and temperature sensors for fire detection, and a mini water pump for water-based suppression. The design aims to demonstrate educational value, affordability, and practicality rather than industrial-scale deployment.

Methodology / Implementation

- **System Architecture & Design:**

We began with a carbon fiber composite quadcopter frame measuring between 550–650 mm to build the drone. This is both durable and lightweight, assisting our drone with longer flight times and stability. Four brushless hobby motors (2212/1000–1200KV) were fitted for quiet and powerful flight, as well as matching ESCs (Electronic Speed Controllers). We selected the propellers (two sets, 10x4.5" or 12x4.5") for balance. We employed a student-friendly flight controller (APM 2.8 or Matek F405 Mini) compatible with ArduPilot or Betaflight. Power is provided by a 3S/4S 5000 mAh lithium battery,

which has a longer lifespan based on what the drone is carrying. Control messages are transmitted using a 6–8 channel FlySky FS-i6 transmitter.

- **Sensor Integration & Functional Parts:**

For firefighting, we incorporated an IR flame sensor, a temperature sensor (MLX90614), and an ESP32-CAM for real-time video. A mini submersible water pump and tank (or fire extinguisher ball mount) perform the suppression task. The same camera module and high-resolution imaging in bridge inspection mode assist in recording multiple vantage points. GPS and IMU sensors make the drone stable and provide complete coverage of the inspection area. This sensor unit makes our drone capable of performing the flight and extinguishing task efficiently

- **IOT Control & Integration:**

The drone is controlled by a smart ESP32 microcontroller. It communicates via Wi-Fi / BT so we can operate it and monitor live video/data with a laptop or smartphone. The system also has simple obstacle avoidance and automated flight paths for safe missions. We can send commands via our remote device to control the drone and monitor its functioning.

- **Reliability & Testing:**

To render the drone reliable, we utilized backup power routing, double-checked wiring and tested every component for high reliability. We conducted real-world tests—such as mock fires and bridge scans—and improved what we had learned from them, such as balancing flight and how the sensors function.

COMPONENT	SPECIFICATION
Flame Detector (IR)	Detects flame in 0-1000 ppm
Temp Sensor (MLX90614)	- 70 to +382°C
ESP32-CAM	2 MP, Wi-Fi, basic vision
High-res Camera	8 MP
GPS Module	<3m accuracy

IMU (Motion sensor)	6-axis
Mini Water Pump	12V, 2-4 L/min

Expected Outcomes / Results

The project aims to design and build a semi-autonomous firefighting drone that can detect and respond to fire-related anomalies in real time. The drone will not rely on heavy onboard computing; instead, it will stream sensor data and video feed to a connected laptop, where monitoring and control take place.

Key outcomes include:

- **Anomaly Detection through Infrared Sensors:** The drone will continuously scan its environment for sudden heat signatures or flame sources. Detected anomalies will trigger alerts on the laptop interface, enabling the pilot to take immediate action.
- **Remote-Controlled Firefighting System:** A human pilot can manually navigate and control the drone to direct water spray through a pipe connected to a ground tank.
- **IoT-Based Monitoring:** Sensor readings such as temperature, and fire status will be transmitted wirelessly, Timestamp of detection, allowing for remote supervision and logging.
- **Low-Cost and Sustainable Design:** The setup uses easily available components like IR sensors, 1000KV motors, Matek F405/Flight Controller, and a 433 MHz relay system to achieve firefighting capabilities at a fraction of commercial costs.
- **Safety and Real-Time Responsiveness:** Human supervision combined with AI-assisted detection minimizes false positives and ensures fast response with reduced human risk.
- **Sustainability and scalability:** The design focuses on using affordable, reusable, and energy-efficient components — aligning with sustainable development goals like SDG 9 (Industry, Innovation, and Infrastructure) and SDG 13 (Climate Action)

This approach demonstrates how AI, IoT, and manual control can complement each other to create a safe, efficient, and affordable firefighting solution.

Future Scope

The prototype opens doors to several possibilities for improvement and real-world application:

1. **Autonomous Navigation:** Future versions could include GPS, LIDAR, and obstacle-avoidance sensors for full autonomy.
2. **Thermal Imaging:** Adding a thermal camera could help the drone “see through smoke” and locate fire sources more accurately.
3. **Edge AI Processing:** Upgrading to controllers like ESP32-S3 or Jetson Nano could enable faster, onboard fire recognition without cloud dependency.
4. **Multi-Drone Coordination:** In large-scale emergencies, a swarm of drones could work together for faster coverage and control.
5. **Smart Analytics:** Using IoT data to predict high-risk fire zones and send preventive alerts before incidents occur.
6. **Industrial Integration:** Companies in logistics, manufacturing, and electronics (like Samsung or Tata) could use this system as part of their safety automation framework to prevent costly fire-related incidents.

References / Bibliography

1. <https://www.mdpi.com/2076-3417/11/1/363>
 2. <https://academic.oup.com/iti/article/doi/10.1093/iti/liad023/7370942>
 3. <https://hsetdata.com/index.php/ojs/article/view/302>
 4. <https://www.mdpi.com/1424-8220/22/9/3310>
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